Android Build Process on Dev Kit

11 Steps — From Host Setup to Running Android

Step 01

PREPARE BUILD HOST

Main Tasks:
- Install Ubuntu 64-bit
- Install repo/Python/ Java/GCC
- Set env variables

Meaning: Stable Linux machine for AOSP compilation

Step 02

GET ANDROID SOURCE

Main Tasks:
- repo init
- repo sync
- Download AOSP (~200 GB)

Meaning: Sync full Android source

Step 03

CONFIGURE BOARD

Main Tasks:
- Edit BoardConfig.mk
- Select ARM/ARM64 arch
- Set kernel & partition params

Meaning: Define hardware-specific build rules

Step 04

BUILD BOOTLOADER

Main Tasks:
- Build U-Boot / Android Bootloader
- Create bootloader.img
- Verify boot

Meaning: First program on power-on — loads kernel

Step 05

BUILD LINUX KERNEL

Main Tasks:
- Configure kernel for SoC
- Build drivers (display/ camera/WiFi/audio)
- Output: zImage

Meaning: OS core — controls hardware & resources

Step 06

BUILD HAL + VENDOR

Main Tasks:
- Build HAL (HIDL/AIDL)
- Build vendor modules
- Validate drivers

Meaning: Bridges hardware and Android framework

Step 07

BUILD ANDROID SYSTEM

Main Tasks:
- source envsetup.sh → lunch → make -j$(nproc)
- Build ART, framework, apps

Meaning: Full Android OS compiled — system.img, boot.img

Step 08

GENERATE IMAGES

Main Tasks:
- Package system.img, boot.img, vendor.img
- product.img
- Sparse if needed

Meaning: Final flashable image files ready

Step 09

FLASH IMAGES TO KIT

Main Tasks:
- Connect via USB
- Flash via fastboot / U-Boot
- Verify flash logs

Meaning: Write images to device eMMC/SD storage

Step 10

BOOT & TEST ANDROID

Main Tasks:
- Power on board
- Test display/camera/audio/ sensors
- Check adb logcat

Meaning: Verify Android boots and all features work

Step 11

DEBUG / FIX / REBUILD

Main Tasks:
- Check dmesg & logcat
- Fix kernel/driver/HAL bugs
- Rebuild & re-flash

Meaning: Iterate until system is stable and reliable

Android Core Architecture

Official 6-Layer Software Stack — Android Open Source Project (AOSP)

⑥ SYSTEM APPS & USER APPS

Home | Contacts | Phone | Browser | Camera | Settings | + User Apps

Meaning: Pre-installed system apps and third-party user apps — the visible layer users interact with

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⑤ JAVA API FRAMEWORK (Application Framework)

Activity Manager | Window Manager | Content Providers | View System | Package Manager

Meaning: High-level APIs for app developers — manages app lifecycle, UI, data sharing, and system services

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④ ANDROID RUNTIME (ART)

AOT Compilation | JIT Compilation | DEX Execution | Garbage Collection | Core Java Libraries

Meaning: Runs Android apps — compiles DEX bytecode using Ahead-of-Time (AOT) + Just-in-Time (JIT)

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③ NATIVE C/C++ LIBRARIES

OpenGL ES | SQLite | SSL/TLS | Media Framework | WebKit | libc

Meaning: Core C/C++ libraries providing low-level system functions used by the framework and ART

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② HARDWARE ABSTRACTION LAYER (HAL)

Camera HAL | Audio HAL | Bluetooth HAL | Sensors HAL | GPS HAL

Meaning: Standard interfaces between hardware and Android framework — hardware-independent abstraction

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① LINUX KERNEL

Device Drivers | Memory Management | Process Management | Network Stack | Power Management

Meaning: Foundation of the entire stack — manages hardware directly and provides core OS services

Each layer only communicates with the layer directly below it — ensuring modularity and hardware independence.